

THE MORPHOSYNTAX OF POSSESSIVE PRONOUNS, REFLEXIVES AND SILENT NOMINAL IN MONGOLIAN¹

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1 Introduction

Focusing on the morpho-syntax interface of Mongolian pronouns, this paper analyzes the formation of possessive pronouns, the possessive-reflexive properties of the pronoun *öör-in* and the suffix *-aa*, and the syntactic status of the suffix *-h*, identified as silent nominals, within the framework of Distributed Morphology. The underlying assumption is that pronouns are not lexical items but rather feature clusters that are driven by syntactic operations such as fusion and realized by vocabulary insertion, following Davis (2023). It is proposed that the lexicon contains phi-features [phi], the possessive feature [poss] and the reflexive feature [ref]; what we call pronouns are formed in syntax by clustering appropriate features onto a single terminal node. Genitive pronominal forms in Mongolian thus come as the result of inserting morphemes such as *mini* into the fused outcome of two bare D heads, each with [phi] and [poss]. The other forms, including the accusative, dative, instrumental, ablative and comitative, result from the insertion of appropriate morphemes, which are transparent. The pronoun *öör-in* is formed by clustering [poss] and [ref].

This paper is organized as follows. Section 2 outlines empirical facts about Mongolian pronouns. Section 3 outlines the portmanteau-formation analysis of possessive pronouns, which was provided by Davis (2023), and extends it to reflexive pronouns. Section 4 applies this analysis to Mongolian possessive pronouns, reflexives and silent nominals. Section 5 discusses the typological implications of the proposed analysis. Section 6 concludes this paper.

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2 Empirical Facts and Questions

Mongolian pronouns have three basic forms: the nominative, the genitive and the ADIAC (for accusative, dative, instrumental, ablative and comitative). The nominative is unmarked and the others are marked. Morphologically, the ADIAC is totally transparent and regular, whereas the genitive is not.

Table 1. Basic forms of pronouns in Mongolian (plural and honorific forms left out)

Person	Nominative	Genitive	Accusative/Dative/Instrumental/Ablative/Comitative
First	<i>bi</i>	<i>mini</i>	<i>nam-ig/-d/-aar/-aas/-tai</i> ²
Second	<i>či</i>	<i>čini</i>	<i>čam-ig/-d/-aar/-aas/-tai</i>
Third	<i>ter</i>	<i>tüüne</i>	<i>tüün-ig/-d/-eer/-ees/-tei</i>

Possessive pronouns have four forms, where possessivity and independency are all regularly marked.

Table 2. Basic forms of possessive pronouns in Mongolian (informal dialectal forms are left out)

Person	Dependent		Independent	
	Non-reflexive	Reflexive	Non-reflexive	Reflexive
First	<i>mini</i>	<i>(mini-)öör-in</i>	<i>mini-h</i>	<i>(mini-)öör-in-h</i>
Second	<i>čini</i>	<i>(čini-)öör-in</i>	<i>čini-h</i>	<i>(čini-)öör-in-h</i>
Third	<i>tüüne</i>	<i>(tüüne-)öör-in</i>	<i>tüüne-h</i>	<i>(tüüne-)öör-in-h</i>

Reflexivity is expressed by a dedicated suffix *-aa* (and its allomorphs *-ee*, *-oo* and *-öö*) in addition to the possessive impersonal pronoun *öör* ‘self’. Interestingly, *-aa*, normally called the Reflexive(-Possessive) suffix and tagged as RX here, is used in two ways. Typically, it is attached to the reflexive possessive pronoun *öör-in* to form *öör-in-H-öö*,³ where the consonant *H* is inserted before *-öö*. Notably, in Mongolian, combining a syllable ending with the nasal sound *n* and one starting with a vowel demands a consonant between them.⁴

- (1) Sara, **öör-in-H-öö** öröö-g saihan ceberle-rei.
 Sara, **self-GEN-H-RX** room-ACC nicely clean-please
 ‘Sara, clean your room nicely, please.’ (Kullmann and Tserenpil 2015: 262)

Notably, *öör-in-H* here is not the same as *öör-in-h*. That is, *öör-in-H* is not an independent reflexive possessive pronoun; the consonant, notated as the capital letter *-H*, does not function

² A form alternative to *nam* is *nad*, which derives *nad-ig*, *nad-d*, *nad-aar*, *nad-aas* and *nad-tai*, when attached by the case endings.

³ *-Öö* is chosen here, complying with vowel harmony.

⁴ In Cyrillic writing system, the sound /h/, notated as *H* here, is indicated, as in *өөрүүнхөө*. In contrast, in Uighur Mongolian writing system, the classic one, it is not indicated, as in *өөрүүн*. Failing to notice the nature of *H*, Kullmann and Tserenpil (2015: 262) described *öör-in-H-öö* in (1) as the combined form of *öör-in-h* and *-öö*.

the way *-h* in *öör-in-h* or in *mini-h* does. *Öör-in-h* and *mini-h*, being independent pronouns, do not occur before nouns.

- (2) **öör-in-h öröö* ‘Intended meaning: one’s own room’
 **mini-h öröö* ‘Intended meaning: my own room’

Alternatively, RX can also be attached to the possessum. In that case, the pronoun *öör-in* is not obligatory before the noun, as exemplified below.

- (3) Sara **öör-in** öröö-g-**öö** saihan ceberle-sen.
 Sara-NOM **self**-GEN room-ACC-**RX** nicely clean-PST
 ‘Sara cleaned her room nicely.’

- (4) Sara öröö-g-**öö** saihan ceberle-sen.
 Sara-NOM room-ACC-**RX** nicely clean-PST
 ‘Sara cleaned her room nicely.’

Importantly, the presence of the genitive pronoun, for example, *tüüne*, will make the phrase sound weird, as shown below.

- (5) ? Sara *tüüne* **öör-in** öröö-g-**öö** saihan ceberle-sen.
 Sara-NOM 3rd-GEN **self**-GEN room-ACC-**RX** nicely clean-PST
 ‘Sara cleaned her room nicely.’

Both dependent and independent pronouns can bear thematic roles and hence arguments,⁵ but only dependent pronouns take nominal complements.⁶

- (6) Sara **mini-g** hergle-sen.
 Sara-NOM 1sg-ACC use-PST
 ‘Sara used mine.’

- (7) Sara **mini-h-ees** yab-san.
 Sara-NOM 1sg-h-ABL go-PST
 ‘Sara left mine (my house).’

Two questions arise, as listed below, and we provide answers by conducting a “portmanteau-formation” analysis of pronouns.

- (8) Why does the genitive have an irregular and fusional morphology compared with that of the ADIAC?
 (9) Why do dependent and independent possessive pronouns interchangeably arguments?

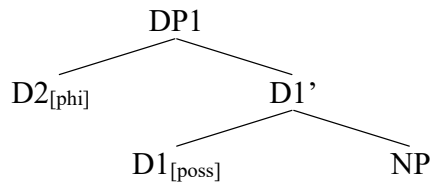
⁵ The use of dependent pronouns as arguments are found particularly in the eastern dialects in Inner Mongolia.

⁶ See Janhunen (2012: 133-141) and Kullmann and Tserenpil (2015: 255-263) for more details.

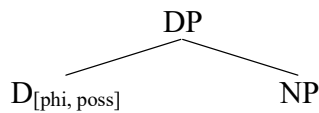
3 Portmanteau Formation of Possessives and Reflexives

Following Davis (2023), possessive pronouns are formed in a “portmanteau” fashion. Possessive forms spell out either a single terminal node D that is the fused outcome of the possessive determiner $D1_{[poss]}$ and $D2_{[phi]}$, which is a bare determiner with phi-features occupying Spec of $D1_{[poss]}$, or multiple terminals that bear appropriate features. Davis (2023) is based on the *bare phrase structure* theory (Chomsky 1995a, b), in which non-projecting heads are equivalent to phrases. Taking *my* as an example, (11) represents the derivation in which fusion (Halle and Marantz 1993; Embick and Marantz 2008) takes place to get the relevant features clustered on a single node before the vocabulary insertion (VI) rule is applied, and (12) represents the derivation in which the VI goes for two adjacent nodes $D1$ and $D2$, instantiating an operation called spanning.

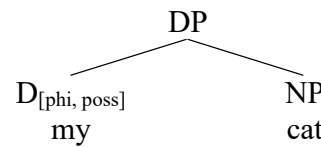
(10) Initial Structure



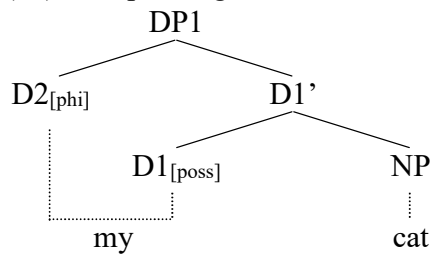
(11) Fusion



Morpho-phonological assignment

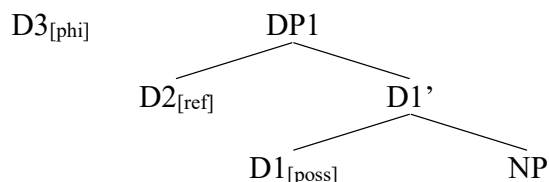


(12) Spanning

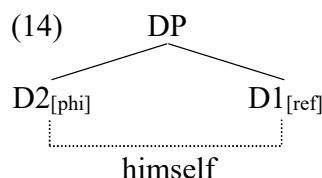


Davis (2023) is concerned only with possessive pronouns, providing no discussion of possessive reflexive pronouns. However, if possessive pronouns are realizations of D with $[phi]$ and $[poss]$, it is reasonable to say that possessive reflexive pronouns are also realizations of D with appropriate features, among which there are $[phi]$ and $[poss]$. We assume that the third feature is $[ref]$, for reflexivity, and that there is another bare D that bears $[ref]$. D with $[poss]$ is the head of the phrase; D with $[ref]$ and D with $[phi]$ are located in Spec of D with $[poss]$ in the initial structure, as shown below.





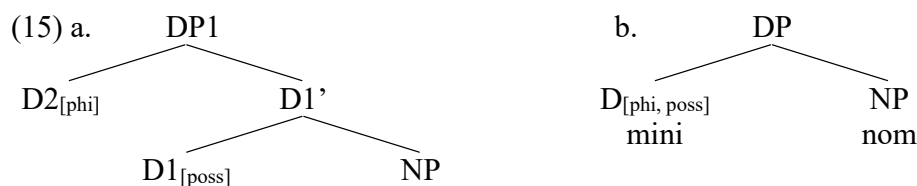
Note that four kinds of sets come out of these three features, namely, [phi, ref], [phi, poss], [ref, poss] and [phi, ref, poss], as inputs for clustering. However, clustering of features as well as fusion of heads is not random. It should be the case that head-head, spec-head or head-comp, but not spec-spec, comp-comp or spec-comp, is subject to fusion. Given this, D3 and D2 do not undergo fusion, with the presence of D1. D_[phi] and D_[ref] may undergo fusion if one of them is the head of the relevant DP, with D_[poss] absent. Alternatively, spanning takes place, where *himself* spells out two terminal nodes with [phi] and [ref].



4 Explaining Possessives, Reflexives and Silent Nominal in Mongolian

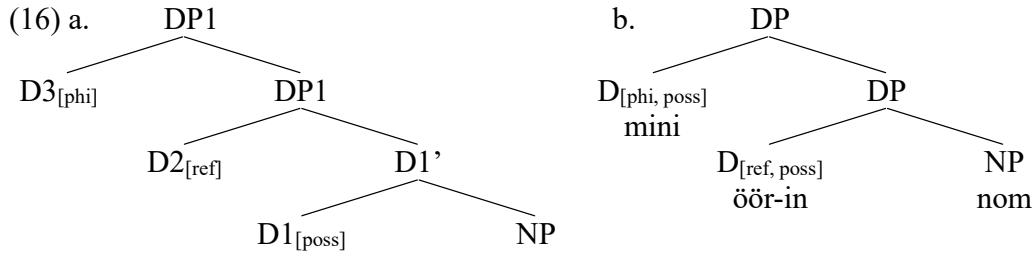
4.1 Possessives

By applying the fusion analysis to *mini* ‘my’ in Mongolian, we obtain the initial structure (15a).



After (15a) is built and before the VI is applied, D2 and D1 undergo fusion, thereby creating a single node, with [phi] and [poss] clustered on it, as in (15b). Then, the VI is applied and *mini* is inserted into D_[phi, poss].

Extending this analysis to reflexive possessives, we argue that their initial structure maximally involves three distinct D heads, each with [phi], a reflexive feature [ref] and [poss], as in (16a). [phi], [ref] and [poss] enter the derivation separately, given that anaphors, with [ref], are often deficient in phi-features (Reuland 2018: 2) and that not all anaphors have [poss]. The complement of D1 is either a full nominal or a silent one with minimal lexical content. Given this structure, the reflexive possessive *mini öör-in* ‘my own’ is syntactically distributed on multiple nodes in its initial structure (16a).



In (16), unlike in (15), both fusion and the VI apply twice. When they apply to $D2_{[ref]}$ and $D1_{[poss]}$, *mini* is inserted into their fused outcome, i.e., $D_{[ref, poss]}$, as in (17b), and when they apply to $D3_{[ref]}$ and $D1_{[poss]}$, *öör-in* ‘own’ is inserted into $D_{[phi, poss]}$. Importantly, after fusion, DP remains a layered projection containing two D heads. $D_{[phi, poss]}$ takes care of personal possession, and $D_{[phi, poss]}$ takes care of reflexive possession. *Öör* lacks phi-features (at least gender and person), which makes it dependent on $D_{[phi]}$. This means that $D_{[phi]}$, whether it has a morphological realization, must be syntactically present for $D_{[poss]}$. Given this, *öör-in* is an impersonal pronoun.

4.2 Reflexives

With *öör-in* present, the possessive *mini* is optionally absent at PF.⁷ This is because possessivity is involved in *öör-in*, and phi-readings are obtained through binding of *mini*, explicit or implicit, by a local subject with [phi]. Recall that in addition to *öör*, we have the suffix *-aa* (and its allomorphs *-ee*, *-oo* and *-öö*) as a reflexive marker, notated as **RX**, in Mongolian. A salient property of **RX** is that it is never attached to the subject (NP1); it is attached to either a possessor (NP3) or a possessum (NP2).

(17) [TP NP1_i-NOM ... [vP ... NP3_k-GEN-**RX** NP2_j-ACC ...] ...]

Assuming that NP1 possesses NP3, which in turn possesses NP2, we obtain the structure in (17), in which NP3 is not coreferential with NP1 and therefore **RX** is attached to it, the possessum of NP1. When coreferentiality is obtained between NP1 and NP3, *öör* is obligatorily chosen for NP3 and is attached by **RX**, as in (18). Alternatively, **RX** may be attached to NP2, the possessum of NP1, as in (19). Interestingly, *öör* is optionally absent at PF. When it is absent, **RX** is necessarily attached to NP2, as in (20).

(18) [TP NP1_i-NOM ... [vP ... NP3_i-GEN-**RX** NP2_j-ACC ...] ...]


(19) [TP NP1_i-NOM ... [vP ... NP3_i-GEN NP2_j-ACC-**RX** ...] ...]


(20) [TP NP1_i-NOM ... [vP ... NP3_i-GEN NP2_j-ACC-**RX** ...] ...]

⁷ *Mini* is mostly absent and some speakers marginally accept the case with it present.

↑
(*öör*)

Another important property of RX is that it is never attached to a possessum with two non-identical possessors. As exemplified in (21), RX is out in the case that the subject of the relative clause is not coreferential with the subject of the main clause. That is, RX is incompatible with switch reference.⁸

- (21) *Baatar bagš-in-H-aa zaa-san hičeel-ø-ee marta-san.
 Baatar-NOM teacher-GEN-H-RX teach-NPST lesson-ACC-**RX** forget-PST
 ‘Baatar forgot the lesson that his teacher taught.’

This switch-reference-resisting property resembles the binding property in which anaphors resist rebinding. Notably, the nominative-resisting property of RX resembles another binding property in which *SELF* anaphors are never nominative. That is, the reflexive markers *SELF* and RX are attached only to non-nominative elements that resist rebinding.

We assume with Reuland (2020) that binding is valuation of [phi]; [phi] of a local subject, a DP in Spec of vP, values [phi] on D, which enters the derivation unvalued.⁹ Under the portmanteau analysis, [ref] is contingent on binding. That is, [ref] is evoked by [phi]-valuation. Supposing that there is no [phi]-valuation, reflexivity would not be obtained.

- (22) [TP SBJ_[phi] ... [vP [DP ... D_[phi] ... D_[ref] ... D_[poss] ... [NP ...] ...] ...]

With this said, coreferentiality as a special kind of dependence goes between the subject and D with [phi], which is “reflected” by the reflexive marker, for example, *self* in anaphors and *own*.

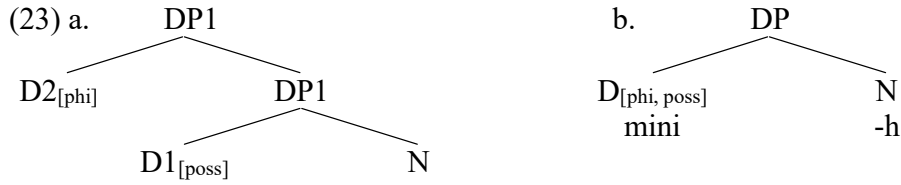
Assuming that RPR is a kind of binding, it is reasonable to say that RX is the marker of reflexivity involved in RPR. However, given that RX as a suffix must be attached to a visible host, it may be attached to the possessed noun when the possessive *öör* is not present at the PF branch of the syntactic derivation. Importantly, *öör* is also a marker of reflexivity by virtue of being a reflexive pronoun.

4.3 Silent Nominal

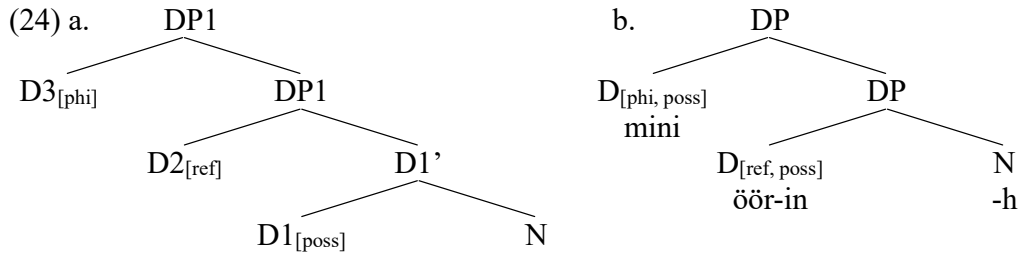
What we refer to as a silent nominal is N rather than NP, which is selected by D with [poss]. The silent nominal involved in, for example, *mini-h* ‘mine’, also lacking [phi], is not a full nominal and therefore is realized not as a lexical word but as a suffix. The VI rule chooses *-h* for it. Ultimately, *mini-h* is formed by combining *mini* and *-h*.

⁸ See Anisman (2021) for detailed discussion on switch reference in Mongolian.

⁹ Kratzer (2009: 216) argued that binding is feature transmission, which starts out from a local verbal head such as v. Unlike Kratzer (2009), Reuland (2020) holds that binding as feature valuation starts out from T. With details left out, what suffices for present purposes is to identify binding as a kind of Agree or feature valuation.



With $D1_{[ref]}$ absent from (23a), *mini-h* is not coreferential with a local subject. When coreferentiality is assigned to *mini* and a local subject, *mini-h*, which functions as an anaphor, has an initial structure as in (24a), in which $D1_{[poss]}$ fuses into $D3_{[phi]}$ and $D2_{[ref]}$, respectively, and the fused outcomes are assigned morphemes *mini* and *öör-in*, as in (24b), the resulting structure.



The silent nominal lacks not only $[phi]$ but also a maximum lexical content. This means that it, analogous to English *one*, is a kind of indefinite pronoun, according to Baker's (2003: 95ff, 127ff) criterion.

5 Typological Implications

The proposed analysis has at least two typological implications. First, English *own*, which has been regarded as a possessive adjective but is functionally equivalent to the unlexicalized form *self's*, is in fact a reflexive possessive pronoun, instantiating the constellation of *SELF* ($D_{[ref]}$) and *'S* ($D_{[poss]}$). This connects to the putative ambiguous reading of sentences such as (25).

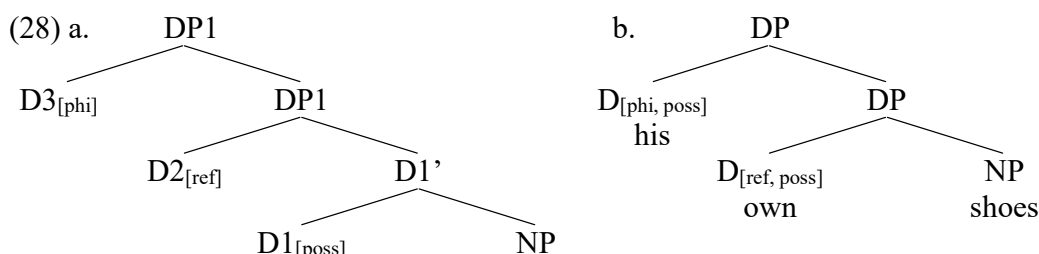
(25) John_i loathes his_{i/j} shoes.

It has been claimed that possessive pronouns in English, for example, *his* are lexically ambiguous between possessive pronouns and reflexive pronouns (Truswell 2014: 226). However, such a claim remains stipulative due to the lack of sound reasoning and empirical evidence. Under the proposed analysis, when *his*, the portmanteau form of *HE* and *'S*, refers back to *John*, the sentence involves an implicit *own* and is subject to Condition A, not to Condition B, with no violation of it.

(26) John_i loathes his_j shoes. (subject to Condition B)

(27) John_i loathes his_i **own** shoes. (subject to Condition A)

Own, as a reflexive possessive pronoun,¹⁰ is derived as follows. [Ref] on D1 and [poss] on D2 are clustered, which gives rise to the single terminal node D as the fused outcome of D1 and D2. This process is followed by the VI, which chooses the morpheme *own* for D, much like the case of *öör-in* in Mongolian.



When [ref] is not present in the derivation, there is no $D_{[ref, poss]}$, hence the failure of the VI to insert *own* as a reflexive possessive pronoun, which leads to sentences such as (26), not (27). It then follows that English does have a reflexive possessive pronoun, namely, *own* and strings such as *his own* are reflexive possessive pronouns that are functionally equivalent to unlexicalized forms such as *himself's* per se.

Second, the silent-nominal analysis accounts for why the same form is employed for both dependent and independent pronouns in some languages, e.g., *watashi-no* ‘mine’ in Japanese and *wo-de* ‘mine’ in Chinese, and why English *yours* and *hers* contain an additional *s* in addition to the genitive form *your* and *her*.

7 Conclusion

This paper applied the “portmanteau-formation” analysis put forward by Davis (2023) in explaining morpho-syntactic properties of Mongolian possessive pronouns and extended it to explain reflexive pronouns in Mongolian as well as English, where binding of reflexives was identified as valuation of [phi] features, which entails the evoking of a reflexivity feature.

The answers to our earlier two questions in (8-9) are as follows. The morphological irregularity of the pronouns is attributed to the fact that their syntactic structure, involving [poss], is richer than that of the ADIAC forms, but their morphology is more compact. The interchangeability of dependent and independent possessives as arguments is accounted for by claiming that they share the same structure with the silent nominal as in (23b), but dependent possessives come as a result of dropping *-h*, the exponent of the silent nominal, for phonological reasons, leading to its silence. When they are not arguments, dependent possessives are assigned the structure in (16b).

A consequence of the proposed analysis is that the blurredness of head-phrase distinction under the bare-phrase spirit leads to the syntactic transparency and morphological compactness

¹⁰ We take reflexive possessive pronouns to be distinct from anaphors on the basis that anaphors bear referentiality and/or [phi], which reflexive possessive pronouns do not. That is, reflexive possessive pronouns can be understood as “anaphors” deprived of their referentiality.

of possessive pronouns. Silent nominals and their equivalents of various types, if any, in other languages are Roots rather than categorizing (mostly nominalizing) heads.

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